

Temporal Pixel Masking Under Calibrated UVC Screen Capture: A Conventional-OCR Feasibility Study

FIRST A. AUTHOR¹ [PLACEHOLDER: Author information to be filled]

¹[PLACEHOLDER: Affiliation and address to be filled]

Corresponding author: [PLACEHOLDER: Corresponding author email].

[PLACEHOLDER: Funding acknowledgment to be filled.] This manuscript includes a planned minimal-risk human-subject study. No participant data have been collected. Recruitment will begin only after a written approval, exemption, or waiver determination by the competent institutional or departmental authority; the planned safeguards are described in Section V-F.

ABSTRACT Screen content can be silently captured and processed by optical character recognition (OCR) or vision-language models (VLMs). Temporal pixel masking offers a possible display-side intervention, but its defensible protection window under physical capture remains unclear. This paper does not claim general screen-photography protection. Instead, it asks whether temporal display profiles can create a measurable single-frame window for fixed-exposure USB Video Class (UVC) pipelines that store isolated photographs for conventional OCR. The method partitions each frame into rapidly alternating complementary subframes, so that a calibrated short exposure records only a sparse pixel subset while a viewer may integrate multiple slots. The physical experiment does not include a luminance-matched static control, so it cannot isolate temporal sparsity from duty-cycle dimming, static overlays, panel response, or exposure. We evaluated one UVC camera across 9 distance/angle configurations and 10,575 captures. Under calibrated short exposure, best-of-engine conventional-OCR character recovery fell from 94.1% unprotected to 15.1% for the readability-priority deployed profile and 5.0% for the high-suppression profile, with exact match falling to 0.4% and 0%. A temporal mean over 150 camera frames, nominally 2.5 s at 60 fps, recovered 71.1% and 47.9%, respectively. At 0.5 m under high-suppression short exposure, Qwen3-VL exactly recovered 28/36 captures from 12 items. These results provide controlled physical evidence that temporal pixel masking can materially degrade unattended conventional-OCR batch collection from isolated fixed-exposure UVC frames. They also show that sustained video aggregation and modern VLMs collapse this narrow protection window, while cross-device behavior and a profile combining validated usability with stronger suppression remain unresolved.

INDEX TERMS Adversarial perturbation, camera capture, display privacy, OCR, persistence of vision, screen-camera capture, temporal masking, visual eavesdropping

I. INTRODUCTION

SCREENS are ubiquitous in offices, banks, hospitals, and conference rooms, and so are capture devices: smartphones, smart glasses [1] (e.g., Ray-Ban Meta), and miniature cameras. A single photograph of a screen suffices to extract text via OCR or identify interface elements through object detection. This *visual eavesdropping* is covert: it leaves no network traces and requires no physical contact with the target device. Controlled experiments and field studies confirm that such attacks are common, succeed at high rates, and frequently go unnoticed [2], [3]. Although these devices motivate our work, the physical evidence presented

here covers only a single UVC webcam; reproducibility on smartphones, smart glasses, and other image signal processor (ISP) pipelines is left to future work.

A. EXISTING APPROACHES AND THEIR LIMITATIONS

Current screen-privacy solutions each occupy a limited niche. Spatial, software, watermarking, and temporal approaches address different viewing geometries or attack stages, but none directly evaluates on-axis short-exposure capture of static sensitive text followed by machine recognition; §II gives the detailed comparison. This leaves a concrete gap between restricting observation and reducing machine recovery

from a physical snapshot.

The difficulty is that degrading information for the camera also tends to degrade it for the legitimate viewer. We ask whether display temporal structure can widen this gap under short-exposure capture.

B. CORE IDEA

Our approach exploits the gap between human temporal integration and camera exposure windows. Each frame is decomposed into n randomly complementary subframes that alternate at high speed. When the exposure is shorter than or close to one subframe duration, the camera records only an incomplete pixel subset, whereas the human observer integrates content across subframes (Fig. 1). Because camera exposure is not instantaneous sampling, long exposure or video reconstruction covering a full cycle can re-sum the complementary subframes; we therefore make no irrecoverability guarantee for arbitrary exposure conditions.

We therefore frame this paper as a controlled measurement study rather than a deployment claim. It asks how much temporal pixel masking reduces conventional OCR recovery from isolated calibrated short-exposure UVC captures, and where that reduction fails. The concrete operational case is an instrumented or surveillance-like UVC collection pipeline in which each target is represented by one fixed-exposure image and processed later by conventional OCR. It is not ordinary smartphone photography, smart-glasses capture, or a targeted operator who can keep recording, adjust exposure, or invoke a VLM. Across the 9 tested geometries, temporal masking reduced conventional OCR character recovery from 94.1% to 15.1% for deployed and 5.0% for high-suppression. Excluding one underexposed geometry raises the protected estimates to 16.7% and 5.6%, so high-suppression remains near but does not meet its strict $< 5\%$ target. The defensible claim is a measured operating window for this narrow UVC condition, not a general anti-camera setting.

The boundary evidence is equally central to the contribution. Ambient illuminance was not recorded, and the calibrated UVC settings cannot be ranked against typical smartphone auto-exposure. The evaluated temporal-mean attack averaged 150 short-exposure frames, nominally 2.5 s at 60 fps; shorter video durations were not evaluated. Among three tested VLM endpoints, Qwen3-VL exactly recovered 28/36 captures from the 12-item pool at 0.5 m under high-suppression short exposure. The separate content-type analysis shows that character recovery concentrates in large-font short snippets (§V-C).

C. CONTRIBUTIONS

The main contributions of this paper are:

- 1) **Controlled profile-level physical measurement:** Building on prior temporal masking ideas, we implement pixel-level complementary subframes with profile-dependent stripe/glyph overlays and a GPU rendering prototype. Using one eMeet SmartCam S600 USB webcam, we aggregate 10,575 captures across

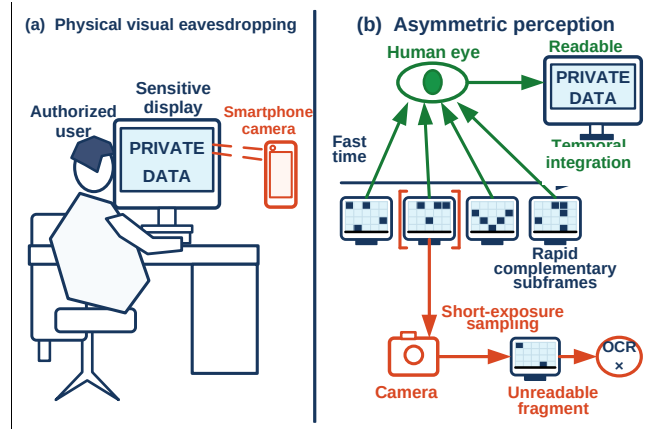


FIGURE 1. Threat scenario and operating boundary of temporal pixel masking. (a) A nearby camera can capture sensitive on-screen content without accessing the host system. (b) Human vision integrates complementary subframes, while a short camera exposure can intersect only part of a display cycle and leave OCR with an incomplete fragment. The camera path is conceptual: it does not assume zero-duration sampling or a fixed human integration time, and long exposure or multi-frame reconstruction can recover much more information.

3 distances and 3 angles. The core calibrated short-exposure subset yields 15.1% character recovery (0.4% exact) for the deployed profile and 5.0% (0% exact) for high-suppression, compared with 94.1% (65.7% exact) unprotected. This establishes a narrow measurement window for unattended conventional-OCR collection from one isolated fixed-exposure UVC frame per target, not default smartphone photography or a general anti-camera setting.

- 2) **Cluster-aware evidence and exposure sensitivity:** We report capture-level descriptive intervals and a content-cluster paired contrast analysis. Matching by profile, attack mode, content item, position, and repeat index, unprotected short exposure exceeds deployed by 78.0 percentage points (95% content-cluster CI [75.5, 80.3]) and exceeds high-suppression by 89.1 percentage points ([85.0, 92.4]) across 324 matched units and 12 content clusters. Excluding the underexposed $d0.5/15^\circ$ position leaves the profile-level conclusions unchanged while moving high-suppression above its strict $< 5\%$ condition-mean target.
- 3) **Measured integration and VLM boundaries:** We quantify where the single-frame window collapses. Per-sample strongest digital attack recovery reaches 95.2%. A temporal mean over 150 short-exposure camera frames recovers 71.1% for deployed and 47.9% for high-suppression; excluding the underexposed geometry raises these values to 79.7% and 53.9%. In the on-axis VLM probe, Qwen3-VL exactly recovers 28/36 captures from 12 items at 0.5 m under high-suppression short exposure. These results motivate a profile-level evidence hierarchy: deployed is the readability-priority candidate but weaker under integration and sensitive-token attacks, while high-suppression reduces recovery

further but lacks user-acceptability evidence.

The remainder of this paper is organized as follows. Section II reviews prior work, Section III defines the threat model, and Section IV presents the design. Section V reports the primary OCR evidence, VLM and integration boundaries, and the deferred usability protocol. Sections VI and VII synthesize the evidence hierarchy and conclude the paper. Detection/tracking diagnostics are separated into supplementary material because they do not extend the primary OCR claim.

II. RELATED WORK

A. SCREEN VISUAL EAVESDROPPING THREATS

Remote screen reading attacks (telescopes, telephoto lenses) [4], deep-learning reconstruction of HDMI electromagnetic leakage [5], covert capture by smart glasses [1], and indirect eavesdropping through reflections and refractions [6] have all been documented. A controlled experiment by the Ponemon Institute found that 91% of visual hacking attempts succeed within 15 minutes [2]. Eiband et al. [3] surveyed 174 participants on the prevalence of shoulder surfing and the coping strategies users adopt. Tang and Shin [7] proposed Eye-Shield, which uses the front-facing camera to detect bystanders and blur sensitive regions in real time, though it relies on user-side hardware and provides only localized masking. Our work focuses on display-side mitigation that requires no attacker-side cooperation, without claiming prevention of information acquisition under all exposure and reconstruction conditions.

B. DISPLAY PRIVACY PROTECTION

Physical privacy filters restrict viewing angles but cannot block an on-axis camera. Chen et al. proposed HideScreen, a display-side shoulder-surfing defense that suppresses low-frequency visual cues so screen content blends with the background beyond a designed viewing distance [8]; it targets human shoulder surfing in the spatial domain rather than short-exposure camera capture or machine recognition. Zhu et al. [9] proposed LiShield, which modulates smart LED illumination to interfere with rolling-shutter cameras, but the scheme acts on ambient lighting rather than displayed content. Visual cryptography splits a secret image into shares that combine upon overlay [10], inspiring temporal decomposition ideas, though its original formulation targets printed media. Digital watermarking [11] can embed tracking identifiers, including screen-shooting-resilient variants [12], but provides post-hoc attribution rather than pre-emptive blocking. Software-based digital rights management (DRM) prevents digital screenshots but not physical photography.

Screen-camera visible light communication (VLC) [13]–[15] similarly exploits human temporal integration to embed information imperceptible to the eye yet decodable by cameras. Our method is VLC's mirror image: VLC aims to let cameras *read* hidden information, whereas we aim to reduce machine readability of displayed content in constrained short-exposure capture.

Yerazunis and Carbone [16] first proposed temporal image masking for display privacy. Wu and Zhai [17] systematized temporal psychovisual modulation, and Gao et al. [18] applied it to information-security displays. Zhang et al. [19] proposed Kaleido, which re-encodes video frames via multi-frame decomposition to exploit the channel difference between display-eye and display-camera paths, making films “viewable but not recordable.” Rainbow [20] addresses a related but distinct physical-world piracy setting: it modulates ambient illumination to corrupt rolling-shutter photographs of objects such as artwork, rather than temporally decomposing screen content. These schemes evaluate continuous recording or illumination attacks using imaging and subjective-quality metrics that are not interchangeable with the OCR, detection, and VLM recovery rates used here. None directly evaluate short-exposure single-frame capture of static screen text coupled with machine recognition. Mechanistically, Kaleido operates in continuous color space with chrominance-complementary frames and luminance variation, relying on perceptually complete mixing within a cycle; our method uses randomized discrete pixel-level partitioning, and the high-suppression profile deliberately abandons strict completeness to resist integration attacks (§IV-E). Lim [21] used persistence of vision to generate anti-screen-capture keypads in a browser, where a single screenshot captures only partial patterns. Table 1 situates the present study among representative spatial, illumination, and temporal schemes rather than treating any one baseline as a complete substitute.

The contribution is not temporal partitioning in isolation. It is the system-level evaluation of a pixel-level randomized implementation against machine reading from physical captures: 10,575 real images across 9 geometries establish the single-frame conventional-OCR operating niche, while integration, sensitive-token, and VLM experiments identify exactly where that niche ends. Detection, tracking, and digital usability proxies remain supplementary diagnostics.

C. ADVERSARIAL PERTURBATIONS AND PHYSICAL-WORLD ROBUSTNESS

Digital adversarial perturbations [22]–[24] and physical-world adversarial patches [25]–[27] are surveyed by Wei et al. [28]. Athalye et al. [29] achieved the first cross-viewpoint-robust 3D physical adversarial objects via Expectation over Transformation (EoT), though their approach still requires white-box access to the target model. A large digital-to-physical robustness gap persists: pixel-level digital perturbations are weakened considerably after camera sampling and JPEG compression [25]. Physical adversarial patches exhibit cross-viewpoint robustness but require perturbation patterns pre-generated for specific classifiers [26], [27], with limited transferability to unseen models [30]. The screen-camera link also introduces moiré artifacts, which Niu et al. [31] demonstrated can themselves act as adversarial perturbations. The base mechanism partitions content temporally at the display layer rather than optimizing a model-specific spatial perturbation. Enhanced profiles additionally add noise and

TABLE 1. Scope Comparison with Representative Display and Physical-Scene Privacy Schemes. Entries Describe Different Threats and Evaluation Targets; They Are Not Matched-Parameter Performance Claims.

Scheme	Protected medium	Target threat	Mechanism	Reported evaluation
HideScreen [8]	Mobile display	Human shoulder surfing beyond a designed distance	Spatial suppression of low-frequency cues	Human-viewing distance/readability
LiShield [9]	Physical scene	Rolling-shutter camera capture	Modulated ambient LED illumination	Captured-image interference
Kaleido [19]	Displayed video	Continuous-recording piracy	Perceptually complete chrominance/luminance re-encoding	Recorded-video PSNR/structural similarity (SSIM) and subjective quality
Rainbow [20]	Physical objects	Mobile-camera piracy	Ambient-illumination modulation	Rolling-shutter image degradation
Lim [21]	Browser keypad	Single digital screen capture	Persistence-of-vision partial patterns	Keypad capture demonstration
This work	Static screen text	Short-exposure physical capture followed by machine recognition	Randomized pixel partitioning; high-suppression profile breaks completeness	OCR/VLM recovery and integration boundaries

stripe/glyph overlays, whose physical transfer is evaluated empirically rather than assumed. Physical photography is inherently a sampling process, and the defense is embedded in the display itself.

III. THREAT MODEL

A. PROTECTED ASSETS

Sensitive text and visual information displayed on screen: passwords, financial data, personally identifiable information (PII), source code, confidential documents, and similar content.

B. ATTACKER CAPABILITIES

Attack capabilities do not form a strict monotonic hierarchy but are jointly determined by exposure, geometry, temporal sampling, and post-processing capabilities. Results are reported across the following scope:

- **Primary scope:** Instrumented short-exposure snapshots and per-frame recognition by conventional OCR attackers. Camera exposure time is shorter than or close to one subframe duration; the attacker may choose among OCR engines but does not possess observations covering a full display cycle. This scope models bulk or unattended collection pipelines that store isolated fixed-exposure UVC photographs for OCR, not targeted capture by an operator willing to hold the camera on the screen. Reproducing this condition outside the laboratory requires a camera or application that exposes manual shutter control and some calibration effort. It should therefore be read as a manual-exposure UVC pipeline model, not as a validated model of ordinary auto-exposure phone snapshots. The experiments use fixed 3.91 ms calibration on one S600 pipeline, while ambient illuminance was not recorded. This is a narrow single-frame condition, not the strongest rational attacker.
- **Practical failure boundary:** Long exposure, video temporal averaging after continuous recording, off-axis capture, phase search and channel reconstruction given a

known period, object detection/association tasks, and end-to-end recognition by commercial vision-language models (§V-C). The evaluated video attack averages 150 short-exposure frames, nominally 2.5 s at 60 fps. It demonstrates a practical sustained-recording bypass but does not establish recovery from an arbitrary four-frame burst. We quantify these risks but interpret high-suppression results only as mitigation, not as a blocking guarantee.

- **Not yet adequately covered:** Additional camera/display combinations, precise rolling-shutter row alignment, real-world temporal super-resolution, learning-based reconstructors trained on display sequences, and long-term adaptive attacks.

C. ATTACKER KNOWLEDGE

The attacker *knows* the temporal masking algorithm, the sub-frame count n , and candidate period parameters, but does *not know* the current cycle's mask seed or phase starting point. The CSPRNG provides per-cycle mask randomization to prevent fixed-pattern learning and cross-cycle correlation; it does not provide a cryptographic guarantee against full-cycle averaging attacks. Once an attacker obtains and registers a complete cycle, an unknown seed cannot prevent linear reconstruction. More fundamentally, the method's security against short-exposure capture rests on disrupting the binarization and segmentation stages of conventional OCR pipelines, not on an information-theoretic impossibility of recovery from sparse pixels. End-to-end recognizers such as VLMs bypass binarization via visual encoding and language priors, and can recover substantial text from the same sparse fragments (§V-C).

D. PROTECTION GOALS

We define tiered design goals under short-exposure conditions by protection profile: the high-suppression profile targets conventional OCR character recovery $< 5\%$ with exact-match = 0%; the readability-priority deployed profile targets

$\geq 80\%$ relative reduction in character recovery with exact-match $< 1\%$. Changes in object-detection mean average precision at 0.50 intersection over union (mAP50) explore task-transfer risk beyond text and are not primary pass criteria. These thresholds guide parameter selection and profile trade-offs but are not formal security guarantees against all attack modes; VLM results are reported separately as security boundaries (§V-C) and do not enter pass/fail determination. The ΔE_{00} and SSIM values are digital reconstruction proxies rather than human-perception evidence. Because the planned user study has not been conducted, we make no “imperceptible to the human eye” claim.

IV. SYSTEM DESIGN

A. TEMPORAL SUBFRAME DECOMPOSITION

Given a normalized original frame $\mathbf{I} \in [0, 1]^{H \times W \times C}$, we generate binary masks $\mathbf{M}_1, \dots, \mathbf{M}_n$ satisfying $\sum_{k=1}^n \mathbf{M}_k = \mathbf{1}$, and set $\mathbf{I}_k = \mathbf{I} \odot \mathbf{M}_k$. This yields:

- **Digital-domain completeness:** $\sum_{k=1}^n \mathbf{I}_k = \mathbf{I}$;
- **Mutual exclusivity:** each pixel is illuminated in exactly one of the n time slots.

A distinction between “summation” and “temporal averaging” is necessary. On an ideal linear display with equal slot durations and equal peak luminance, the average radiance over one cycle is $\mathbf{I}/n$, not $\mathbf{I}$. Digital-domain completeness means subframes can be re-summed; it does not imply that real display luminance has been losslessly preserved.

This distinction is also an experimental limitation. The physical capture study did not include a luminance-matched static dimming control, a matched noise-off version of the enhanced profiles, or photometric panel timing measurements. The observed profile-level reductions therefore cannot be attributed to temporal sparsity alone; duty-cycle luminance, static stripe/glyph overlays, panel response, camera exposure, and ISP processing remain entangled.

Masks are generated using a ChaCha20 [32] CSPRNG and the Fisher–Yates shuffle [33], randomly assigning each pixel to one subframe. The implementation uses a chi-square slot-count check ($\alpha = 0.01$, $\text{df} = n-1 = 3$) as a software sanity check and re-samples the seed upon failure (up to 5 retries). It is not evidence of cryptographic strength: for $n = 4$, each assignment already comes from an unbiased 2-bit distribution, and a correctly implemented generator is expected to trigger the nominal false-rejection rate. Random seeds reduce fixed patterns and cross-cycle correlation but do not change the fact that a full cycle can be linearly summed. Fig. 2 illustrates the complete pipeline from original frame to protected display.

B. COMPLEMENTARY ADVERSARIAL NOISE

We overlay adversarial noise $\mathbf{N}_1, \dots, \mathbf{N}_n$ on the subframes, satisfying:

$$\sum_{k=1}^n \mathbf{N}_k = \mathbf{0} \quad (1)$$

Equation (1) holds strictly only in the ideal linear domain prior to summation, clipping, and the display transfer function. Prior work has shown that small text-image perturbations can significantly mislead OCR systems [34], providing a basis for OCR-oriented noise design. Noise directions are generated by fast gradient sign method (FGSM)-style [22] proxy gradients with an amplitude bound of $\varepsilon = 8/255$, then decomposed complementarily in the temporal domain. Because displays cannot emit negative radiance, the implementation uses a pedestal level and numerical clipping. Consequently, after gamma correction, quantization, clipping, and camera imaging, the noise is not guaranteed to cancel physically.

Component role: The random dot-pattern mask is the primary temporal mechanism. Complementary noise is an optional profile component generated in the ideal linear domain; stripe/glyph overlays and partial inversion are nonlinear components used by the enhanced profiles below. The composite profiles were fixed before the physical ablation. Retaining noise in their definitions preserves one-to-one correspondence with the archived captures; removing it after observing the ablation would define an untested profile. Its inclusion is therefore a traceability choice, not evidence of benefit or optimality. The physical results in §V-B show that noise is nonmonotonic, and no matched noise-off versions of the deployed or high-suppression enhanced profiles were captured. The manuscript therefore does not recommend complementary noise as a default deployment component; matched noise-off enhanced profiles are a required follow-up before profile optimization.

C. LUMINANCE COMPENSATION

Each pixel is illuminated only once per n -slot cycle. If the display could boost light output by a factor of n in the linear-light domain during the active slot, the cycle-averaged luminance would approach the original. The current proof of concept (PoC) implements neither per-slot dynamic backlight coordination nor photometric verification. All luminance alignment, CIEDE2000 ΔE_{00} [35], and SSIM results therefore come from digital integration/normalization models and serve only for cross-configuration comparison, not as evidence of real-panel optical equivalence or user comfort.

D. TIMING AND BANDWIDTH

For a basic cycle containing only n complementary subframes, taking 60 Hz as the minimum cycle rate requires $f_r \geq 60n$; at $n = 4$ this corresponds to 240 Hz. If the cycle additionally includes one inversion frame, the requirement becomes $f_r \geq 60(n+1)$, and at 240 Hz the actual cycle rate drops to only 48 Hz. Thus, 240 Hz meets the basic four-subframe configuration but not the inversion-frame configuration’s 60 Hz cycle target. Human flicker perception also depends on luminance, contrast, retinal eccentricity, spatial content, and eye movements [36], [37]; nominal cycle rate alone cannot predict comfort.

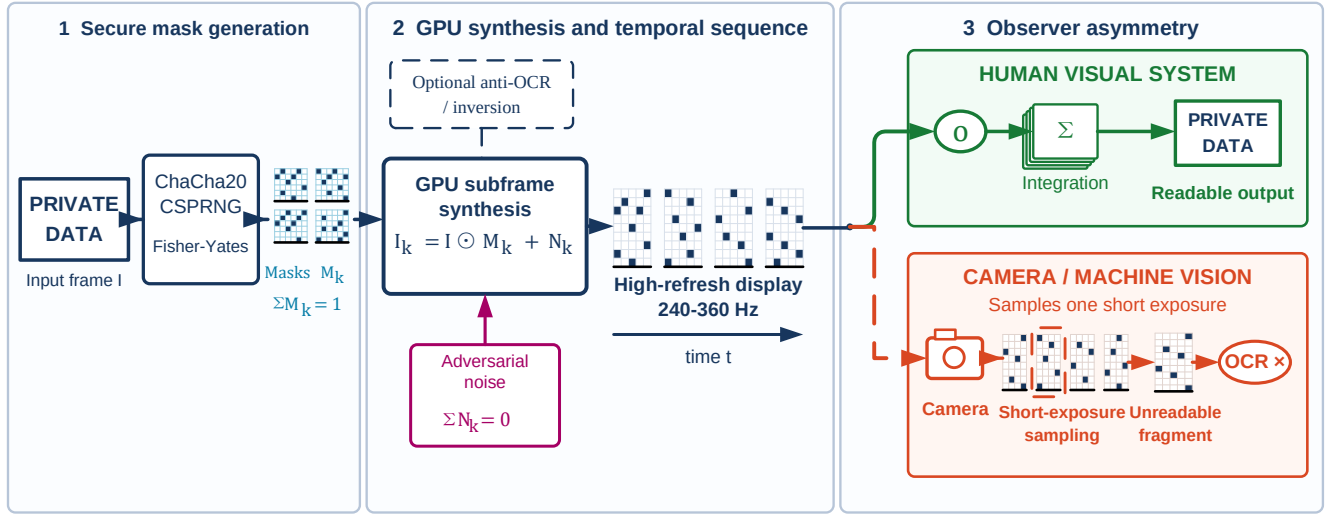


FIGURE 2. Temporal pixel masking system pipeline from source content to protected display. The input frame is decomposed into CSPRNG-assigned complementary pixel masks; optional complementary OCR-oriented noise, profile-dependent stripe/glyph overlays, and an optional partial inversion frame are then inserted before high-refresh playback. A human observer is expected to integrate multiple slots into a readable view, whereas a short-exposure camera may capture only a sparse subframe. Long exposure and video temporal averaging can accumulate multiple slots and are therefore evaluated separately as boundary attacks rather than assumed away.

Whether an observer integrates complementary subframes into a perceptually complete image depends on the temporal contrast sensitivity function (TCSF). Critical flicker fusion (CFF) is not a universal cutoff: Davis et al. [37] found sensitivity near conventional rates for uniform modulation but visibility above 500 Hz when high-frequency spatial edges were present. The 48 Hz full-cycle rate of the inversion configuration may therefore produce visible flicker for some observers, and the 60 Hz basic configuration cannot be assumed flicker-free. Only the planned user study can determine whether either configuration supports comfortable temporal integration.

E. HIGH-SUPPRESSION PROFILE

The basic linear temporal mask leaks under full-cycle temporal averaging (see §V-E2). To address this, we introduce nonlinear enhanced profiles. Unless otherwise stated, profiles in this subsection use $n = 4$ base temporal masking with complementary adversarial noise enabled. Table 2 summarizes all profile configurations; the “Mask + noise” row in Table 4 is the base condition without stripe/glyph anti-OCR overlay.

- **Strong (anti-OCR) profile:** base mask+noise with mask cell size of 1 pixel, stripe width of 10 pixels, stripe amplitude $\alpha_s = 0.10$, and glyph amplitude $\alpha_g = 0.12$, without an inversion frame.
- **Deployed profile:** the Strong profile plus an $\alpha = 0.2$ inversion frame, preselected as the readability-priority composite configuration before the physical ablation. It is not an empirically optimized component combination.
- **High-suppression profile:** base mask+noise with mask cell size increased to 2 pixels, stripe width of 6 pixels, stripe and glyph amplitudes of 0.42 and 0.55 respectively, plus an $\alpha = 0.2$ inversion frame.

The nonlinear profiles depart from strict completeness ($\sum I_k \neq I$), producing visible stripes and grain. This trade-off is the opposite of Kaleido-class temporal re-encoding: the latter relies on perceptually complete mixing within a cycle to preserve viewing experience, meaning full-cycle integration inherently reconstructs an approximation of the original, consistent with its anti-piracy goal. In the static-content capture threat addressed here, completeness is precisely the entry point for integration attacks; the high-suppression profile therefore makes breaking strict completeness an active design trade-off, accepting visible stripes and grain in exchange for lower recovery under full-cycle integration. Its security benefit and unmeasured readability cost must be reported separately.

We use the paper-facing name “high-suppression” because it describes the profile’s relative position in Table 2 without implying that a capture-resistance target has been met. The archived identifier remains `capture_hardened` (with legacy tag `v1m`) for reproducibility; §V-B reports the target miss and residual leakage.

F. PARTIAL INVERSION FRAME

A long-exposure attack occurs when the exposure window covers a full display cycle ($n = 4$ at 240 Hz yields a basic cycle of approximately 16.7 ms; with an inversion frame the $n + 1$ slot cycle is approximately 20.8 ms; §V-A reports measured long-exposure times of 31.25 and 125 ms). Such attacks approximate the original image through sensor integration. A direct countermeasure inserts a **partial inversion frame** $\alpha \cdot (255 - I)$ after each n -subframe cycle, introducing a luminance component complementary to the original into the long-exposure integral and reducing net signal contrast. Higher α may reduce long-exposure recovery, but also alters

TABLE 2. Profile Composition Summary ($n = 4$ for All Profiles)

Profile	Mask	Noise	Cell (px)	Stripe width (px)	Stripe α_s	Glyph α_g	Inv. α
Mask only	✓	–	1	–	–	–	–
Mask + noise	✓	✓	1	–	–	–	–
Strong	✓	✓	1	10	0.10	0.12	–
Deployed	✓	✓	1	10	0.10	0.12	0.2
High-suppression	✓	✓	2	6	0.42	0.55	0.2

TABLE 3. Inversion Frame Strength α Ablation on the Fixed Strong Overlay ($\alpha_s = 0.10$, $\alpha_g = 0.12$): Real-Capture Best-of-Engine Character Recovery (Mean and 95% Bootstrap CI, Five Account/Code/CET6 Items Pooled Across 9 Geometric Conditions; $N = 135$ per Setting for Long Exposure, $N = 45$ for Video Temporal Averaging). This Five-Item Subset Differs from the 12-Item Deployed Rows in Table 4.

α	Long-exp. char (%)	Video avg. char (%)
0.0	68.6 [63.1, 73.9]	72.7 [63.3, 81.1]
0.2	67.0 [61.3, 72.7]	69.8 [59.9, 78.7]
0.3	72.0 [66.2, 76.9]	64.3 [54.0, 73.6]
0.5	61.7 [55.6, 67.7]	66.4 [56.5, 75.2]
1.0	18.1 [14.0, 22.0]	28.4 [20.5, 36.9]

the luminance and contrast of the digital integrated view.

A real-capture inversion strength ablation (Table 3; pipeline per §V-B) shows that weak inversion does not meaningfully reduce long-exposure recovery: $\alpha = 0$ yields 68.6% and $\alpha = 0.2$ yields 67.0% (best-of-engine), with 95% confidence intervals overlapping across $\alpha \in [0, 0.5]$. The ablation holds the Strong overlay fixed ($\alpha_s = 0.10$, $\alpha_g = 0.12$) and varies only inversion strength, using the five-item account/code/CET6 subset rather than the 12-item main corpus; its $\alpha = 0.2$ value ($N = 135$) is therefore not a repeated estimate of the deployed row's 60.9% ($N = 324$). Only near-full inversion ($\alpha = 1.0$) reduces recovery to 18.1%, at the cost of severely degrading the digital integrated view. Under the 150-frame video temporal mean, weak inversion similarly maintains 64–73% recovery. Weak inversion alone cannot resist integration attacks. The deployed choice of $\alpha = 0.2$ is a modeled readability-priority design decision, not a user-study-validated optimum.

Each display cycle thus contains $n + 1$ time slots. At $n = 4$ and 240 Hz the cycle rate is 48 Hz, below the 60 Hz design target and potentially producing perceptible flicker; no fixed “flicker-free threshold” applies universally across viewing conditions.

G. IMPLEMENTATION SCOPE

The current PoC operates at the application/algorithm layer, using Python and GPU rendering to generate display sequences. It does not include OS driver-level injection (DXGI / KMS-DRM / CoreDisplay) or per-slot dynamic backlight control. We therefore validate the display sequence and its captured results, without claiming production-level system integration.

V. EXPERIMENTAL EVALUATION

A. EXPERIMENTAL SETUP

Hardware platform: Experiments ran on a desktop workstation with an Intel Core i7-8700K (3.70 GHz), 32 GB RAM, and an NVIDIA TITAN Xp (12 GB VRAM). The display is an AOC 24G51Z (23.8-inch Fast IPS panel, quantum-dot (QD) backlight, edge-lit LED without local dimming, RGB stripe subpixels), driven at 1920×1080 with a nominal refresh rate of 240 Hz. The manufacturer specifies 1 ms gray-to-gray (GtG) response time; at 240 Hz each subframe slot lasts approximately 4.17 ms, so the GtG transition occupies roughly 24% of one slot. Slower gray-level transitions (particularly dark-to-light on IPS) may take 2–5 ms, causing partial temporal overlap (ghosting) between adjacent subframes. The measured real-capture results already include these panel-response effects; mechanistic narrative should therefore be read as “camera samples a short exposure that may span partial transitions across subframes” rather than “camera captures exactly one discrete subframe.” This refresh rate satisfies only the basic $n = 4$ subframe cycle at $f_r/n = 60$ Hz. With an additional inversion frame per cycle, the actual cycle frequency is $240/(n+1) = 48$ Hz. Thus, the visual comfort of the inversion configuration cannot be inferred from “240 Hz high refresh rate” alone and still requires user-study validation.

Capture device: All physical captures use a single eMeet SmartCam S600 USB webcam (UVC standard driver). The playback canvas was 1920×1080 ; camera output underwent perspective correction and content cropping before recognition. Short-exposure and video-frame exposure times were 0.49 or 3.91 ms, long-exposure times were 31.25 or 125 ms, and video was captured at 60 fps. These values are discrete DirectShow/UVC $\log_2$ exposure-control steps recorded by the calibration and capture metadata: -11 and -8 for short/video frames, and -5 and -3 for long exposure. The common final runs use 3.91 and 31.25 ms; 0.49 and 125 ms occur in the disclosed early d0.5/15° run. They define instrument settings, not representative smartphone shutter times.

The camera was placed at 0.5 m, 1 m, and 1.5 m from the display, rotated to 0° , 15° , and 30° at each distance, yielding 9 geometric configurations. Ambient illuminance was not recorded. The manufacturer specifies an 8-megapixel 4K image sensor and 1080p/60 fps output [38], but does not identify a Sony sensor; we therefore make no sensor-model claim. The 60-fps frame period is 16.67 ms, but rolling-shutter readout time and phase relative to the 4.17 ms display slots were not measured. Actual panel output and application-

level slot timing were not verified photometrically. Row-level mixing and panel transitions may help explain why real-capture short-exposure recovery (15.1%) exceeds simulated single-subframe recovery (0–2.6%), but the current data do not isolate either mechanism. Results apply to this UVC link and should not be extrapolated to smartphones, smart glasses, or other camera ISPs.

Test content: At each of the 9 geometric positions, the real-capture OCR experiment used a fixed set of 12 content items: 11 selected synthetic items and one CET6 (College English Test Band 6) document (portrait 900×1280). Three captures per item give 36 captures per profile–attack condition at each position and $12 \times 3 \times 9 = 324$ across all positions before the disclosed deployed-profile repeats. By contrast, the software-simulated OCR experiment used the complete 120-sample synthetic corpus (12 content templates $\times$ 10 variants, mapped to 9 category labels covering passwords, credentials, code, tables, paragraphs, URLs, etc.). Real-capture short-exposure OCR is the primary feasibility evidence. Detection/tracking diagnostics use separate COCO and MOT subsets and are reported only in the supplementary material because their metrics and experimental design do not carry the central OCR claim.

Evaluation metrics: OCR character recovery is defined as $\max(0, 1 - \text{CER})$, where CER denotes Character Error Rate. Exact match is determined after whitespace removal and case-insensitive comparison of the full text. Sensitive-token recovery counts the proportion of reference-text digits, account numbers, URL/path strings, and key-like fields that are recovered in full. Leak rate is the proportion of samples with character recovery $\geq 20\%$, an arbitrary working threshold chosen as a round-number proxy for “non-trivial partial recovery”; no claim is made that 19% recovery is safe. To supplement this binary indicator, the per-capture recovery distribution is characterized by percentiles: for deployed short exposure, $P50 = 8.3\%$, $P75 = 16.7\%$, $P90 = 43.0\%$, $P95 = 60.9\%$, and $P99 = 94.5\%$ ($N=459$); for high-suppression, the corresponding values are 1.6%, 8.3%, 14.1%, 18.7%, and 22.2% ($N=324$). The deployed short-exposure P95 happens to equal the deployed long-exposure mean in Table 4 (both 60.9%); they are different statistics over different attack conditions. The heavy right tail indicates that a small fraction of close-range, large-font captures yield near-complete recovery. Section V-F defines the separate usability protocol and metrics.

OCR aggregation and uncertainty: For each capture, we take the maximum recovery across three engines to model an attacker choosing the best available recognizer. Table 4 reports descriptive 95% bootstrap intervals with 2,000 percentile resamples, using captures as the resampling unit and a fixed seed of 20260612. These intervals summarize variation within the archived corpus and setup; repeated measurements of the same content and geometry mean they are not independent population intervals. For inference-oriented profile comparisons, we additionally construct paired contrasts by matching profile, attack mode, content item, position, and available repeat index, then resampling the 12 content items

as clusters. Duplicate deployed captures with the same content/position/repeat key are averaged before pairing rather than counted as independent matches. Under this matching contract, unprotected short exposure exceeds deployed by 78.0 percentage points (95% content-cluster CI [75.5, 80.3]) and exceeds high-suppression by 89.1 percentage points ([85.0, 92.4]) across 324 matched units. Mask-only exceeds deployed by only 3.0 percentage points ([−0.5, 6.5]), so the current archive does not establish that every deployed-profile component independently improves protection.

Exposure sensitivity: The d0.5/15° position used 0.49 ms and produced visibly underexposed protected frames. Excluding it raises deployed short recovery from 15.1% to 16.7% [14.7, 18.8]%, high-suppression short from 5.0% to 5.6% [4.8, 6.4]%, deployed 150-frame video-mean recovery from 71.1% to 79.7% [75.4, 83.8]%, and high-suppression video-mean recovery from 47.9% to 53.9% [48.0, 59.9]%. The paired content-cluster short-exposure contrasts remain large after the exclusion: 76.7 percentage points ([74.3, 79.2]) for unprotected minus deployed and 88.9 percentage points ([85.5, 91.9]) for unprotected minus high-suppression. The 5.6% sensitivity value is short-exposure character recovery; it is unrelated to the 5.6% video exact-match value in Table 4. The pooled 9-position values remain in the main table to avoid post-hoc deletion.

Additional robustness checks: A leave-one-round-out analysis on the five repeated content items gives deployed short-exposure recovery of 16.6% for round 1 and 15.8% for round 2, leaving the profile-level interpretation unchanged. Detection and tracking differences have not undergone significance testing and are reported descriptively in the supplementary material.

Digital reconstruction proxies: ΔE_{00} is the mean CIEDE2000 color difference between the original image and the full-cycle digitally reconstructed image; SSIM is their mean structural similarity. Both are computed in the digital pipeline and are used only for within-paper profile comparison. Neither metric measures real-panel optical output, temporal comfort, or human readability.

B. REAL-CAPTURE OCR EXPERIMENT

The real-capture OCR corpus comprises 10,575 images (1,175 per geometric position), all from the S600 webcam. The corpus mixes component ablations, parameter sweeps, and main attack conditions; it is not a balanced “9 geometries $\times$ 6 main profiles $\times$ 3 attacks” full-factorial design. One asymmetry requires disclosure: the deployed profile performed two capture rounds for 5 of the 12 content items at each position (2 account credentials, 2 code snippets, 1 CET6 document), giving 459/153 short-exposure/video samples rather than 324/108. Both rounds enter the descriptive condition means. For paired contrasts, repeated deployed captures with the same content/position/repeat key are first averaged so that they do not become extra independent matches. The repeat batch contained only short-exposure and video captures, so no second long-exposure batch was acquired and

the deployed long-exposure row remains $N = 324$. Three OCR engines, Surya [39], EasyOCR [40], and Tesseract [41], were applied, producing 31,725 engine-level records. Recent VLMs show strong OCR capabilities [42]; the probes in §V-C confirm that protection against conventional OCR does not transfer to VLM attackers. Table 4 reports core conditions with per-capture best-of-engine; the paired content-cluster contrasts in the text provide the main profile-level comparison. Table 5 gives per-engine short-exposure results. Fig. 3 shows qualitative comparisons; Fig. 4 summarizes recovery rates by profile and attack mode.

Key findings:

- 1) **Short-exposure recovery drops substantially, but the high-suppression target is not met:** Unprotected screen photographs yield 94.1% OCR character recovery (95% CI [93.1, 95.0]; exact 65.7%). The mask-only condition reaches 19.2%, while mask+noise is worse at 25.6% and Strong is 20.7%; the archive therefore does not establish a monotonic benefit from noise or weak overlays in physical short-exposure capture. The deployed profile (Strong + $\alpha = 0.2$ inversion) reaches 15.1% (CI [13.3, 17.1]; exact 0.4%), an end-to-end observation for the preselected readability-priority composite rather than proof that each component helped. In the paired content-cluster analysis, unprotected short exposure exceeds deployed by 78.0 percentage points (95% CI [75.5, 80.3]) across 324 matched units and 12 content clusters. Sensitive-token recovery is nevertheless 24.0%, showing that high-value short fields are less protected than average text. The high-suppression profile yields 5.0% (CI [4.3, 5.8]; exact 0%) in the full pool, and unprotected exceeds it by 89.1 percentage points ([85.0, 92.4]) in the same matched analysis. This does not meet the strict $< 5\%$ character-recovery target, because excluding the underexposed position raises the condition mean to 5.6% (CI [4.8, 6.4]). These results support short-exposure mitigation feasibility within the tested camera, exposure, and geometry range, not attainment of the high-suppression design target.
- 2) **Integration attacks remain the residual frontier; long-exposure reversal is a notable negative result:** Both long exposure and sustained video temporal averaging exploit full-cycle complementarity. The deployed profile still recovers 60.9% char / 36.4% exact under long exposure (sensitive-token recall 96.5%) and 71.1% char / 42.5% exact under the 150-frame video mean. Excluding the underexposed $d0.5/15^\circ$ position raises deployed video recovery to 79.7% (CI [75.4, 83.8]), so the full-pool value understates the practical integration leakage. The divergence between character and sensitive-token recovery reflects the structural properties of high-value fields: digits, account numbers, and URLs are typically short, rendered in large font with a restricted character set, and thus recoverable from degraded images even when surrounding dense

text is not. **Long-exposure reversal:** the unprotected long-exposure baseline is only 47.3% char, lower than the deployed profile's 60.9%. At the sole 125 ms position ($0.5\text{ m}/15^\circ$), unprotected recovery is 1.1% versus 46.3% deployed; the observed clipping of the full-brightness image supports saturation reduction and a shift into the sensor's approximately linear response region as the explanation for this position. Excluding it, the 31.25 ms anomaly remains position-dependent: at 0.5 m the expected direction holds (unprotected 78–82% $>$ deployed 25–55%), whereas several $1.0\text{--}1.5\text{ m}$ positions show deployed recovery 12–59 pp higher. Without matched local image diagnostics, we only *hypothesize* that residual stripe/glyph edges interact with geometry-specific overexposure or contrast compression; the data do not establish this mechanism. The anomaly does not weaken short-exposure conclusions, where deployed is always lower, but it demonstrates that fixed long-exposure parameters produce nonmonotonic effects. Weak inversion ($\alpha = 0.2$) alone is insufficient to resist temporal integration. The weak Strong overlay likewise provides no measurable integration defense: mask-only and Strong are nearly identical under the 150-frame video mean (79.0% versus 78.6%) and differ only modestly under long exposure (67.2% versus 61.6%), with overlapping bootstrap intervals. Measurable real-capture integration reduction appears only at high-suppression amplitudes, and even there leakage remains substantial. The primary validity conclusions are confined to single-frame short exposure; integration attacks are reported as a security boundary.

- 3) **High-suppression profile: reduced integration leakage with unresolved perception cost:** This profile yields 9.3% char (CI [7.9, 10.9]) / 0% exact for long exposure and 47.9% char (CI [41.8, 54.0]) / 5.6% exact (CI [1.9, 10.2]) under the 150-frame video mean. The video leakage is substantial and does not constitute attack closure. The pooled 47.9% masks large inter-position variation: 0.5 m at three angles yields 50.7%/0%/33.8%; 1.0 m yields 61.1–69.4%; 1.5 m yields 48.4–55.5%. The 0% point is the underexposed $0.5\text{ m}/15^\circ$ session discussed in §V-A; excluding it raises video char recovery to 53.9% (CI [48.0, 59.9]) and exact match to 6.3%, confirming that the full-pool result is defense-favorable. Deployment assessment should therefore use worst-case geometry and exposure. The digital pipeline shows lower reconstruction quality for high-suppression, but this is not real-capture or human-subject evidence. The user study protocol (§V-F) excludes this profile, leaving its perceptual cost unquantified.
- 4) **Geometric trends:** Across 9 conditions ($0.5\text{--}1.5\text{ m} \times 0\text{--}30^\circ$), protected short-exposure recovery is consistently below unprotected values. Because the design is unbalanced and no inferential model for geometric factors has been fitted, we do not claim statistically

TABLE 4. Single-UVC Real-Capture OCR Core Conditions (Best-of-Engine; N = Captures per Row; Pooled Across 9 Geometric Conditions). Video Rows Average 150 Short-Exposure Frames per Capture Burst. Repeated Content and Geometry Mean That Captures Are Not Independent Population Samples. The Pool Includes the Underexposed d0.5/15° Position; Exclusion Sensitivities Are Reported in the Text.

Profile	Attack mode	N	Char recovery [95% CI]	Exact match	Sensitive token	Leak rate
Original (unprotected)	Short exposure	324	94.1 [93.1, 95.0]%	65.7%	98.4%	99.4%
	Video (150-frame mean)	108	79.9 [73.5, 86.0]%	61.1%	85.6%	86.1%
	Long exposure	324	47.3 [42.8, 51.9]%	18.8%	95.7%	58.6%
Mask only	Short exposure	324	19.2 [17.0, 21.4]%	0.3%	30.8%	30.6%
	Video (150-frame mean)	108	79.0 [73.0, 84.9]%	48.1%	84.0%	88.0%
	Long exposure	324	67.2 [63.0, 71.5]%	35.5%	94.0%	75.0%
Mask + noise	Short exposure	324	25.6 [22.5, 28.8]%	2.8%	34.4%	36.1%
	Video (150-frame mean)	108	79.2 [73.3, 85.0]%	49.1%	83.9%	88.9%
	Long exposure	324	68.0 [63.7, 72.4]%	39.8%	95.7%	76.2%
Strong (anti-OCR)	Short exposure	324	20.7 [18.2, 23.3]%	0.6%	33.7%	32.4%
	Video (150-frame mean)	108	78.6 [72.5, 84.5]%	50.9%	83.3%	88.9%
	Long exposure	324	61.6 [57.1, 66.1]%	39.8%	95.7%	71.3%
Deployed	Short exposure	459	15.1 [13.3, 17.1]%	0.4%	24.0%	20.5%
	Video (150-frame mean)	153	71.1 [65.7, 76.2]%	42.5%	78.5%	85.0%
	Long exposure	324	60.9 [56.6, 65.6]%	36.4%	96.5%	69.1%
High-suppression	Short exposure	324	5.0 [4.3, 5.8]%	0.0%	6.5%	3.7%
	Video (150-frame mean)	108	47.9 [41.8, 54.0]%	5.6%	45.6%	76.9%
	Long exposure	324	9.3 [7.9, 10.9]%	0.0%	34.9%	14.2%

significant monotonic robustness in distance or angle.

Per-engine results: Table 5 breaks down short-exposure results by engine; Fig. 5 visualizes all six profiles. The three engines differ widely in unprotected recovery (Surya 37.1%, EasyOCR 79.5%, Tesseract 84.3%). Under Strong, recovery ranges from 2.1–16.3%; under deployed, 3.2–11.7%; under high-suppression, 1.8–2.0%. All three engines show reduction, but with only three engine families the finding is not recognizer-agnostic. Best-of-engine takes the maximum across engines per capture, modeling an attacker who picks the best recognizer available in this evaluation. It is a conservative aggregation from the defender’s perspective within the tested engine set, not an upper bound over all available OCR services; optimized or commercial OCR pipelines may recover more text.

The evaluated engines are open-source conventional OCR systems; commercial OCR APIs such as Google Cloud Vision, Azure AI Vision, or Baidu OCR were not tested and may achieve higher recovery. The three-engine best-of-engine result should therefore be read as an attacker-favorable aggregation within a limited engine set, not as an upper bound over available OCR services.

C. REAL-CAPTURE VLM PROBES

To test whether conventional OCR conclusions extend to stronger recognizers, we applied three commercial VLM endpoints exposed through the SiliconFlow API (Qwen3-VL-32B-Instruct [43], Kimi-K2.6, and GLM-4.5V [44]) to the same real-capture photographs. This section is a *boundary probe*: if VLMs recover text where conventional OCR fails, the practical scope of the method is confined to conventional OCR short-exposure mitigation.

The probes cover 0° on-axis at three distances. The 0.5 m

TABLE 5. Real-Capture Short-Exposure OCR: Per-Engine Results (Pooled Across 9 Geometric Conditions)

Profile	Engine	Char (%)	Exact (%)	N
Original	Surya	37.1	22.8	324
	EasyOCR	79.5	27.8	324
	Tesseract	84.3	47.8	324
Mask only	Surya	4.2	0.3	324
	EasyOCR	15.6	0.0	324
	Tesseract	3.2	0.0	324
Mask + noise	Surya	11.0	2.8	324
	EasyOCR	18.1	0.0	324
	Tesseract	4.8	0.0	324
Strong	Surya	6.1	0.6	324
	EasyOCR	16.3	0.0	324
	Tesseract	2.1	0.0	324
Deployed	Surya	3.2	0.4	459
	EasyOCR	11.7	0.0	459
	Tesseract	3.3	0.0	459
High-suppression	Surya	1.8	0.0	324
	EasyOCR	2.0	0.0	324
	Tesseract	1.9	0.0	324

run includes 156 inputs: 36 unprotected short-exposure base-lines, 36 high-suppression short- and long-exposure inputs each, and four aggregate views from 60 fps video (12 segments each). The 1.0 m and 1.5 m runs each include 120 high-suppression inputs: 36 short- and long-exposure inputs each plus four video aggregate types (12 segments each). VLM inputs and the OCR best-of-engine (BoE) reference in Table 6 use the same file batch (3.91 ms short/video-frame exposure and 31.25 ms long exposure). All inputs are raw camera JPEGs without contrast enhancement; pilot tests showed that CLAHE-type enhancement partially reconstructs masked text and is therefore an attack component.

Unprotected - Human eye (integrated)

The quick brown fox jumps over the lazy dog

Unprotected - Short exposure (single frame)

The quick brown fox jumps over the lazy dog

Unprotected - Long exposure

The quick brown fox jumps over the lazy dog

Deployed - Human eye (integrated)

The quick brown fox jumps over the lazy dog

Deployed - Short exposure (single frame)**Deployed - Long exposure**

The quick brown fox jumps over the lazy dog

High-suppression - Human eye (integrated)

The quick brown fox jumps over the lazy dog

High-suppression - Short exposure (single frame)**High-suppression - Long exposure**

FIGURE 3. Real-capture montage. Within each vertically stacked profile group (unprotected, deployed, and high-suppression), panels show, from top to bottom, the digital integrated reconstruction view (for checking full-cycle image consistency), short-exposure capture, and long-exposure capture. Digital integrated views are not direct evidence of human readability or visual comfort.

Of 1,188 model requests, 1,133 returned parseable results: 34 failures at 0.5 m (468 calls; Qwen 2, Kimi 0, GLM 32), 21 at 1.0 m (360 calls; Qwen 4, Kimi 1, GLM 16), and none at 1.5 m (360 calls). Each cell reports effective N and is conditional on a parseable response. GLM failures concentrate on mixed-content (47–54%), digit-string (23–38%), and account-credential (31%) items, suggesting content-dependent rather than random missingness. Therefore, cells with failed calls are descriptive and should not be compared directly across models or distances. For the GLM 0.5 m short-exposure cell ($N_{\text{eff}} = 31$), imputing failures as 0% gives 61.0% char / 44.4% exact, while imputing them as 100% gives 74.9% / 58.3%; the reported result is 70.8% / 51.6%.

We use the failure-free 1.5 m run as primary cross-model evidence; the 1.0 m run enters the short-exposure distance

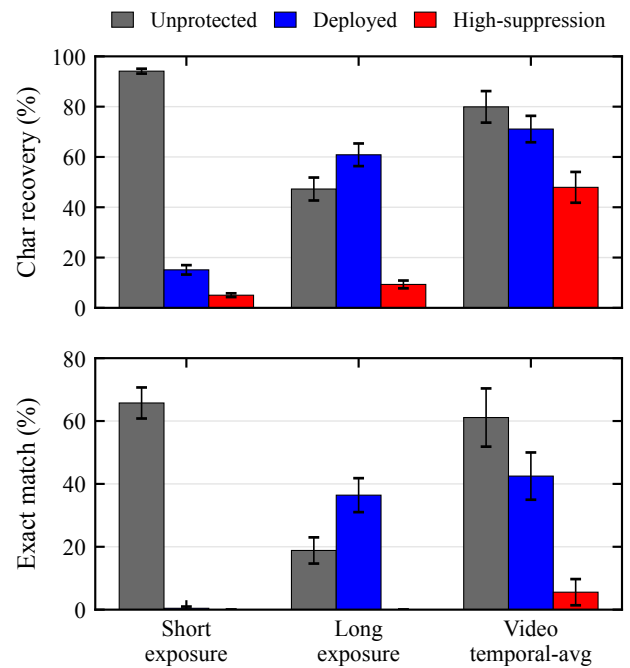


FIGURE 4. Best-of-engine OCR character recovery and exact match for three representative real-capture profiles (unprotected, deployed, and high-suppression) under short exposure, long exposure, and a 150-frame video temporal mean, pooled across the 9 tested distance-angle configurations. Lower bars indicate less recovered text. The long-exposure and video bars expose the integration boundary: the readability-priority deployed profile retains substantial recoverable content even when its short-exposure bar is low. Deployed sample counts include the disclosed five-item repeat and therefore differ from the other profiles.

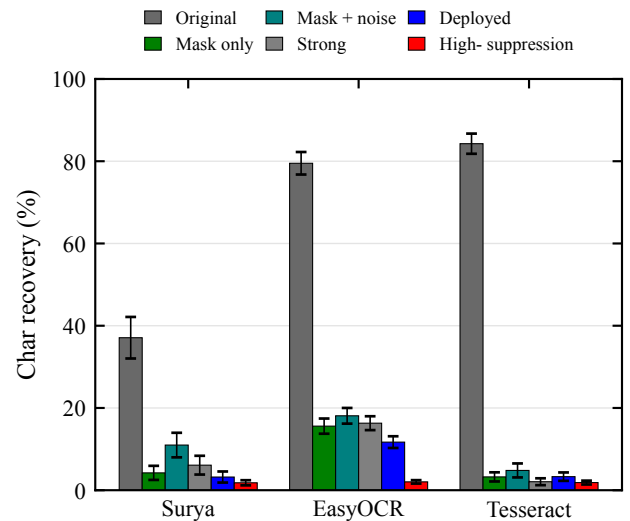


FIGURE 5. Real-capture short-exposure character recovery for three OCR engines across all six profiles in Table 5; error bars are 95% bootstrap intervals resampled by capture. All three engines are suppressed, but with a limited engine family this cannot be claimed as recognizer-agnostic.

sweep as supplementary data. At 1.0 m, Qwen and Kimi long-exposure exact match are 84.4% and 74.3%, and the

successful calls from the three models yield 70.0–83.3% exact match under the 150-frame video mean. GLM long-exposure exact match is only 3.3% at 1.0 m versus 58.3% at 1.5 m, but content-dependent failures preclude a cross-distance consistency claim. For compactness, Table 6 reports only the 150-frame video mean at 0.5 m; the 1.5 m run reports all four aggregate views to show view-selection sensitivity.

All calls used SiliconFlow’s OpenAI-compatible API. The recorded provider identifiers are Qwen/Qwen3-VL-32B-Instruct, Pro/moonshotai/Kimi-K2.6, and zai-org/GLM-4.5V. Qwen3-VL and GLM-4.5V have public technical reports [43], [44]. For Kimi-K2.6, we use the provider endpoint identifier and the archived request metadata as the reproducibility record, rather than making claims about a separately documented upstream public release. The Pro/ prefix in the second identifier is SiliconFlow-specific, and the endpoint accepted image inputs in our archived calls. Decoding temperature is 0 and maximum output is 4,096 tokens. The shared prompt requests only visible text and prohibits guessing; ground truth is not provided. Full prompts, parameters, and per-call outputs are archived. Metrics follow §V-A and are pooled at the capture level. Public CET6 documents may inflate recovery through training-data memorization, so synthetic snippets and exact match provide cleaner discriminative evidence.

0.5 m session selection transparency: The archive preserves an alternative session (real_capture_vlm_d0.5_a0_rerun.json, timestamp 012715) at the same position whose high-suppression short-exposure inputs are nearly all-black, yielding empty transcriptions and 0% char/exact across all three models. Table 6 uses the 141107 (3.91 ms) session because it shares the OCR reference batch and its inputs are not black. These are the only two 0.5 m on-axis VLM sessions, so Qwen’s observed short-exposure exact-match outcome ranges from 0/36 captures in the underexposed session to 28/36 captures in the tabled session, not a stable single-session estimate. The adequate-exposure criterion was applied after inspecting two sessions and is therefore a post-hoc selection rule; preserving both sessions and reporting the range makes the exposure sensitivity explicit. The tabled session is the conservative attacker-favorable case, while the alternative is retained as an underexposure diagnostic.

Table 6 presents the pooled VLM probe results: short-exposure rows show distance variation, while long-exposure and video aggregate rows show integration attack boundaries. Table 7 breaks down 0.5 m high-suppression short-exposure character recovery by content type.

Key findings:

- 1) **VLMs bound rather than extend the claim:** On the same 0.5 m photographs, best-of-engine OCR yields only 8.4% char / 0% exact on the high-suppression short-exposure condition, while Qwen3-VL achieves 91.5% char and exactly recovers 28/36 captures from 12 items. At 1.5 m, the three models retain 33.3–47.2% exact / 61.5–70.0% char. Even with a prompt prohibiting guessing, VLMs recover substantial text from vis-

ible fragments. Within this on-axis, high-suppression, three-distance probe, conventional OCR claims therefore do not transfer to VLM attackers; off-axis VLM capture and other protection profiles were not evaluated.

- 2) **Failure concentrates on large-font short snippets:** At 0.5 m, CET6 full-page dense documents yield 0.4–18.5% VLM character recovery (unprotected baseline 64.4%), whereas credentials, code, and digit strings yield 50–100% (Table 7). At 1.5 m, CET6 dense pages drop to 0% char across all three models, while digit strings and short sentences remain highly recoverable. The pattern matches mask granularity: small-font strokes (~ 1 –2 pixels wide) lose glyph topology when $1 - 1/n$ of pixels are removed. Large-font glyphs retain global contours even through sparse fragments, and VLMs can complete them via visual encoding and language priors [42], [45], bypassing the binarization bottleneck that disrupts conventional OCR.
- 3) **Attack capability varies across models and conditions:** Under 0.5 m long exposure, Qwen3-VL recovers 82.4% exact while Kimi/GLM do not exceed 2.8%; at 1.5 m, long-exposure exact rises to 58.3–91.7%. VLM screen-reading ability depends jointly on model version, distance, exposure, and content; any conclusion about “defeating current VLMs” will expire as models update.
- 4) **Sustained video aggregation is the VLM residual frontier:** At 0.5 m, the 150-frame video mean yields 83.3–87.5% exact / $\geq 91.8\%$ char; the three other 0.5 m aggregate views (single-best, window-mean-best, and max-projection) yield Qwen exact of 66.7%, 83.3%, and 83.3% respectively. These views are all derived from the same 150-frame burst, so they characterize view selection after sustained recording rather than independent short-duration attacks. At 1.5 m, the 150-frame mean yields 58.3–66.7% exact / 88.5–90.3% char. By contrast, 1.5 m single-best-frame yields only 0–8.3% exact, and conventional OCR on the same temporal-mean views stays at or below 8.3% exact (48.4–50.7% char). For VLM attackers, multi-cycle integration (§V-E2) is far more dangerous than selecting a single clear frame, but this experiment does not determine the minimum recording duration needed.

The effectiveness claim must therefore be bounded by content and attacker type: the high-suppression profile protects dense small text against conventional OCR but fails for large-font short snippets (credentials, code, digit strings) when the attacker uses a modern VLM. This negative result repositions the practical scope from “blocking arbitrary machine screen-reading” to “reducing conventional OCR short-exposure batch recovery,” and identifies VLM-aware display protection as an open problem. Countermeasure directions are discussed in §VI-E. The section covers a single camera at 0° on-axis, one profile, three commercial API models, and

TABLE 6. Real-Capture VLM Probes (0° On-Axis; All Rows Except the Original Short Baseline Are High-Suppression). Each VLM Cell Reports Exact Count / Char / *N* (Recoveries / % / Effective Calls); Cells with Failures Are Conditional on Parseable Responses. OCR BoE Shows the Same-Batch Three-Engine Best-of-Engine Reference (Exact / Char, %). The Temporal-Mean Rows Average 150 Short-Exposure Frames; Other Aggregate Views Are Defined in the Text.

Distance	Condition	OCR BoE	Qwen3-VL-32B	Kimi-K2.6	GLM-4.5V
<i>Short-exposure distance sweep</i>					
0.5 m	original short	58.3 / 92.4	30 / 94.5 / 36	30 / 95.6 / 36	19 / 92.0 / 26
0.5 m	short	0.0 / 8.4	28 / 91.5 / 36	17 / 84.7 / 36	16 / 70.8 / 31
1.0 m	short	0.0 / 3.6	20 / 79.1 / 36	14 / 68.6 / 36	9 / 47.0 / 33
1.5 m	short	0.0 / 6.5	17 / 68.9 / 36	12 / 61.5 / 36	13 / 70.0 / 36
<i>Key attack views</i>					
0.5 m	long	0.0 / 2.1	28 / 85.7 / 34	1 / 7.9 / 36	0 / 4.9 / 30
0.5 m	Video (150-frame mean)	8.3 / 50.7	10 / 94.0 / 12	10 / 95.5 / 12	7 / 91.8 / 8
1.5 m	long	0.0 / 13.2	33 / 89.3 / 36	32 / 89.9 / 36	21 / 62.4 / 36
1.5 m	Video (single best)	0.0 / 4.3	1 / 24.8 / 12	0 / 12.7 / 12	0 / 29.8 / 12
1.5 m	Video (150-frame mean)	0.0 / 48.4	8 / 90.3 / 12	8 / 90.3 / 12	7 / 88.5 / 12
1.5 m	Video (window-mean best)	0.0 / 8.6	7 / 89.0 / 12	5 / 84.2 / 12	8 / 87.2 / 12
1.5 m	Video (max projection)	0.0 / 19.0	8 / 90.1 / 12	7 / 89.2 / 12	7 / 88.7 / 12

TABLE 7. 0.5 m High-Suppression Short-Exposure VLM Character Recovery by Content Type. Baseline Column Shows Same-Session Unprotected Original Short Results for Qwen (Kimi/GLM Similar); GLM Had Occasional Call Failures, Parenthesized *N* Shows Effective Count. At 1.5 m, CET6 Dense Pages Yield 0% Char Across All Three Models.

Content type (<i>N</i>)	Baseline (%)	Qwen (%)	Kimi (%)	GLM (%)
CET6 document (3)	64.4	18.5	5.6	0.4
Credentials (6)	97.6	97.6	88.3	77.1 (5)
Code snippets (6)	89.8	94.9	95.8	88.9
Digit strings (6)	100.0	100.0	98.6	50.0 (4)
Mixed content (6)	100.0	100.0	92.5	98.2 (4)
Paragraphs (6)	100.0	99.8	81.7	66.5
English sentences (3)	95.3	95.3	96.9	94.6

only 12 video segments per cell; numbers are descriptive boundary probes, not full cross-geometry or cross-model-version evaluation.

D. SUPPLEMENTARY DETECTION AND TRACKING DIAGNOSTICS

Object-detection and tracking diagnostics are reported in the supplementary material rather than the main evidence chain. They use one real-capture position, a limited COCO subset, still-frame MOT captures, and approximate simulation metrics, so they are diagnostic stress tests but do not extend the primary conventional-OCR claim. The normalized cross-task summary figure is omitted from the main paper because OCR character recovery, mAP, HOTA, and IDF1 are not commensurate measurements.

E. SOFTWARE SIMULATION SUPPLEMENTARY EXPERIMENTS

This section uses software simulation to address two questions: (1) what is the machine readability of a single protected subframe obtained from the display compositor? (2) what is the attacker's recovery boundary under zero camera noise with perfect alignment? Question (1) also characterizes single-frame screenshots: a screenshot tool captures the framebuffer at one instant, corresponding to one protected

TABLE 8. Synthetic Single-Subframe OCR: Three-Engine Character Recovery (120 Samples, $n = 4$, $\varepsilon = 8/255$)

OCR engine	Original frame	Single subframe	Reduction
Tesseract	94.0%	0.0%	94.0%
EasyOCR	94.1%	0.0%	94.1%
Surya	97.0%	2.6%	94.4%

subframe. The 0–2.6% recovery rates below apply to this scenario. The boundary is clear: if malware intercepts the original frame before masking, or screen recording accumulates a full cycle, the method offers no protection. We claim only single-frame-level mitigation, not a general “anti-screenshot” guarantee.

1) Synthetic Single-Subframe OCR

Using Tesseract [41], EasyOCR [40], and Surya [39] on 120 synthetic samples, character recovery drops from 94.0–97.0% to 0–2.6% across three engines (Table 8). The result covers the current corpus, parameters, and engine set; stratified breakdowns can check category trends but are insufficient to prove effectiveness across all languages, layouts, or recognizers.

Fig. 6 visualizes these results. Single-subframe recovery does not exceed 2.6% for any engine, confirming the finding is not driven by one recognizer. Broader generalization requires additional engines and corpora.

2) Theoretical Security Boundary

When an attacker under ideal conditions accumulates frames covering a full cycle, the complementarity can be exploited and complementary noise can cancel in the ideal linear model of (1). The archived digital evaluation recovers 95.2% under the *per-sample strongest attack across the evaluated attack set*; this value is not the pure temporal-mean result. The corresponding pure full-cycle temporal-mean attack recovers 94.3%, with other channel/projection attacks reaching similar values. This is the *fundamental boundary* of an ideal

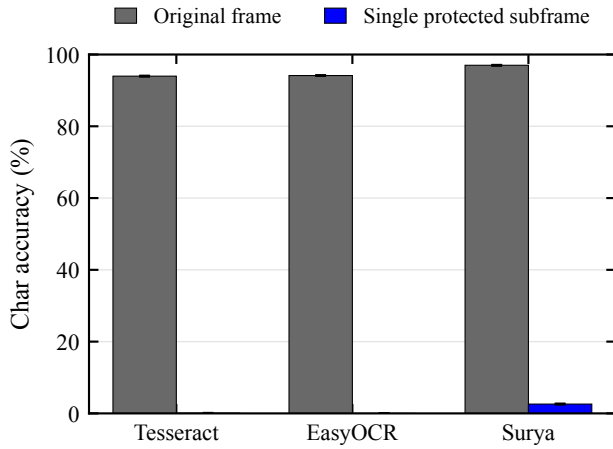


FIGURE 6. Character recovery for three OCR engines on original frames and single protected subframes; error bars are 95% bootstrap intervals resampled by corpus sample.

linear temporal decomposition. The high-suppression profile reduces recovery under the current real-capture integration conditions, but the 150-frame video mean still yields 47.9% pooled character recovery and 53.9% after excluding the underexposed position. It can therefore only be called mitigation, not closure of this boundary. Character recovery need not scale with the illuminated-pixel fraction $1/n$, because recognizer performance also depends on glyph topology, image content, display response, optics, and ISP processing. The current experiments do not provide a formal information-theoretic bound.

Additional software-proxy baseline, parameter-sweep, detection, and tracking diagnostics are preserved outside the main evidence chain. They are not used as main-text evidence because their physical conditions, user-experience costs, and metrics are not matched to the real-capture OCR estimand.

F. USER EXPERIENCE STUDY

To evaluate the impact of the protected display on human readability and visual comfort, we designed a web-based user study.

Equipment and environment protocol: The same 240 Hz display as in §V-A. The formal study version sets a hard measured refresh-rate threshold of ≥ 200 Hz to ensure the fixed $n = 4$ basic subframe cycle rate $f_r/n \geq 50$ Hz; below this threshold participation is blocked rather than silently reducing n . The 144–199 Hz adaptive path is available only for an explicitly labeled demonstration mode, and its sessions are excluded from analysis by default. Before each participant begins, the experimenter confirms fixed display brightness, disabled auto-brightness/power-saving and variable refresh rate, unified browser fullscreen, closed high-load background processes, and an approximate 60 cm viewing distance. Refresh rate is re-measured after any display-mode change.

The browser records actual frame intervals in each trial's requestAnimationFrame loop; if the interval between

adjacent callbacks exceeds $1.5\times$ the expected interval computed from the pre-trial measured refresh rate, a dropped-frame event is logged, along with estimated drop count, drop rate, mean/max frame intervals, observed refresh rate, basic subframe cycle rate, and full cycle rate with inversion frame. This recording reflects only browser presentation cadence and cannot substitute for panel scanning or photometric measurement.

Ethics and safety: The planned study involves minimal-risk human-subjects research, and no participant data have been collected. Before recruitment, the authors will obtain and document a written approval, exemption, or waiver determination from the competent institutional or departmental authority; formal data collection will not begin without that determination. Participants must separately check informed consent and photosensitivity screening: the page explicitly states that the masked display involves rapid temporal modulation, that participants should stop immediately upon experiencing discomfort, eye fatigue, dizziness, nausea, or headache, and that they may withdraw unconditionally at any time. Individuals who self-report a history of photosensitive epilepsy or sensitivity to flicker are not permitted to proceed. The system records consent timestamps and screening pass flags. The implementation applies a conservative $f_r/n \geq 50$ Hz operating floor, but this engineering gate is not an ethical or clinical safety determination. At $n = 4$ on a 240 Hz panel the basic cycle rate is 60 Hz; the inversion frame reduces the full cycle frequency to $f_r/(n + 1) = 48$ Hz, and the implementation separately logs warnings and actual timing. Registration information (student ID, name) is used only for participation management; analysis and publication use only de-identified data. Age and gender are optional demographic fields.

Task design: Each participant first completes one 12-second, unscored unmasked Canvas typing warm-up, views a 10-second, unscored deployed full-mask preview, and then completes an 8-second, unscored typing practice under the same deployed mask. This sequence separates initial stimulus and input adaptation from the four subsequent 20-second scored trials (within-subjects design). On every typing screen, including warm-up, practice, and scored trials, the stimulus Canvas is visible and its player has started before the participant clicks “Start.” Clicking “Start” triggers a 5-second countdown while the input remains disabled; the timed typing interval and input availability begin only when the countdown ends. The same pre-start viewing procedure applies to both control and masked conditions.

The two conditions are each repeated twice. The experimenter's continuously assigned zero-based sequence number k selects ABBA or BAAB order to counterbalance practice and fatigue: (A) **original text** (control), displayed as unmasked Canvas with $n = 1$; (B) **masked text**, using the actual deployment configuration: $n = 4$, mask+noise, strong anti-OCR ($\alpha_s = 0.10$, $\alpha_g = 0.12$, 6 cycles) + weak inversion ($\alpha = 0.2$). The typing order index is $k \bmod 2$, and the subjective-rating Latin-square row is $\lfloor k/2 \rfloor \bmod 6$. Every

consecutive 12 participants therefore cover all 2×6 joint assignments. Formal sessions enforce a uniqueness constraint on k , and the backend recomputes and verifies both order indices.

Both conditions are rendered by the same `render SourceCanvas` path, with identical dark background, light text, canvas size, 23 px bold font, and color. The sole processing difference is the temporal protection pipeline, avoiding polarity and font-weight confounds. Here “cycles” means that the web implementation rotates through six distinct mask, noise, and stripe instances before repeating. This web-playback-specific setting shares α_s and α_g with the real-capture configuration in §IV and avoids repeatedly integrating one fixed overlay pattern.

Scoring no longer compares characters position-by-position but aligns the full transcription against the optimal target prefix using Levenshtein minimum string distance (MSD) [46]; untyped target suffix does not count as error. The alignment yields accuracy, characters per minute (CPM), and words per minute (WPM), along with edit distance, aligned prefix length, and scoring version. First-keystroke latency is defined from the end of the 5-second countdown, when the input is enabled and the scored timer starts, to the first input event that makes the transcription non-empty. Analysis first averages each participant’s two trials per condition, then performs within-subject paired comparison.

Subjective ratings: After the typing trials, participants rate 6 conditions on the three 1–5 Likert items displayed by the interface: readability (1 = difficult to read, 5 = very clear), stability (stored in the `flicker` field; 1 = strong flicker, 5 = no perceptible flicker), and immediate visual comfort (`fatigue`; 1 = very uncomfortable, 5 = very comfortable). Higher values therefore indicate a better outcome on every item. Conditions include the unmasked original anchor, $n \in \{2, 3, 4\}$ mask+noise, $n = 4$ mask-only, and the same $n = 4$ deployed full configuration (mask+noise+strong anti-OCR+weak inversion) used in the typing task. The first three masking levels all achieve ≥ 60 Hz real temporal playback on a 240 Hz panel; the originally planned $n = 8$ would yield 30 Hz and trigger static fallback, so it is not included as a temporal ablation. The six conditions are presented in a 6th-order balanced Latin square; each condition must be viewed for at least 10 seconds before submission is enabled, with viewing start/end times and duration recorded. The unmasked anchor is also an attention/scale-interpretation aid but is not mechanically excluded by a “must rate 5” rule. This study does not include the high-suppression profile; assessment of its visible stripes and grain’s acceptability is left to future work (see §VI).

Sample size and analysis plan: A minimum of $N = 24$ participants is planned, a number within the range required for paired effect sizes of $d_z \approx 0.6$ – 0.8 at 80% power, and exactly covering two rounds of the 12-person joint order assignment. Pre-registered exclusion criteria are debug/demo sessions, incomplete four typing or six rating records, measured refresh rate below 200 Hz, any typing trial with fewer

TABLE 9. Typing Task: Within-Subject Comparison of the Control and Deployed Masked Conditions ($N = XX$; Participant Means Over Two Trials per Condition; ‘ = Masked – Control With Participant-Level Bootstrap 95% CI). Per Metric, a Paired t -Test Is Used When the Pre-Specified Shapiro Test Passes (Effect Size Cohen’s d_z), Otherwise a Wilcoxon Signed-Rank Test (Effect Size Rank-Biserial r_{rb}); Test Statistics Are Reported in the Text

Metric	Control	Masked	Δ [95% CI]
Accuracy (%)	–	–	– [–, –]
WPM	–	–	– [–, –]
CPM	–	–	– [–, –]
Attempted chars	–	–	– [–, –]
First-key latency (ms)	–	–	– [–, –]

TABLE 10. Subjective Ratings for the Six Display Conditions ($N = XX$; 1–5 Likert, Higher Is Better; Mean $\pm$ SD). Stability Is the Inverse-Flicker Item and Comfort the Immediate Visual Comfort Item (SV-F); Friedman and Holm-Corrected Pairwise Wilcoxon Statistics Are Reported in the Text

Condition	Readability	Stability	Comfort
Unmasked anchor	–	–	–
$n=2$ mask+noise	–	–	–
$n=3$ mask+noise	–	–	–
$n=4$ mask+noise	–	–	–
$n=4$ mask-only	–	–	–
Deployed full ($n=4$)	–	–	–

than 5 attempted characters, control mean accuracy below 50%, masked trials not in temporal mode, any trial with observed basic cycle rate below 50 Hz, or minimum-view straight-line ratings. The last criterion is defined as all six rating rows having a recorded viewing duration no greater than 11 seconds (a 1-second tolerance above the 10-second submission gate) and, within every row, identical scores on all three items. WPM, CPM, accuracy, attempted characters, and first-keystroke latency are first averaged within each participant by condition; speed metrics must be interpreted jointly with accuracy to avoid misinterpreting fast input under low accuracy as performance improvement. Paired differences are tested using paired t -tests when the pre-specified Shapiro test is passed, otherwise Wilcoxon signed-rank tests, with effect sizes and participant-level bootstrap 95% confidence intervals reported. Likert data are analyzed per dimension using Friedman tests with Holm-corrected post-hoc pairwise Wilcoxon tests. Analysis scripts generate de-identified participant means, JSON audit reports, and two $\text{\LaTeX}$ tables directly from the formal database `study_formal.db`.

Participants: XX volunteers were recruited; XX sessions met all pre-registered inclusion criteria and entered analysis (XX excluded; per-criterion audit counts accompany the released analysis output). Optional demographics for the included sample: XX of XX reported age (median XX years, range XX–XX) and XX reported gender (XX women, XX men). All formal sessions ran on the 240 Hz laboratory display with a measured refresh rate of at least 200 Hz.

Results: Table 9 and Fig. 7 summarize the typing task. Relative to the control condition, the deployed full configuration changed mean accuracy by XX percentage points

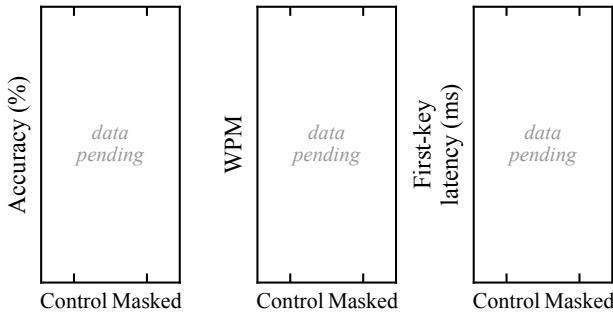


FIGURE 7. Within-subject typing performance under the control and deployed masked conditions: each gray line is one participant (mean of two trials per condition); markers show group means with participant-level bootstrap 95% confidence intervals. Panels show transcription accuracy, words per minute, and first-keystroke latency.

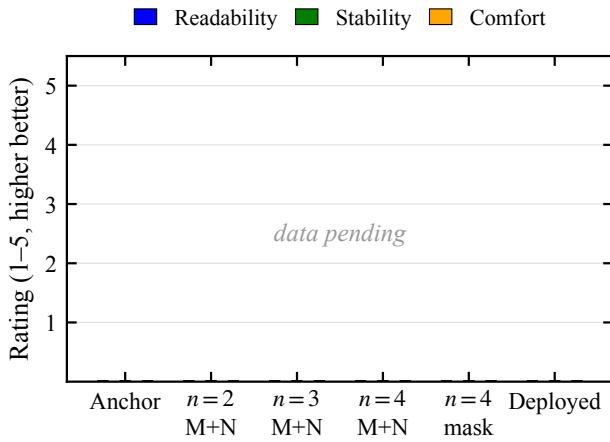


FIGURE 8. Mean 1–5 Likert ratings (error bars: SD) for readability, stability, and immediate visual comfort across the six rated display conditions, ordered from the unmasked anchor to the deployed full configuration. Higher is better on every item.

(95% CI [XX, XX], $p = XX$, effect size XX), WPM by XX (95% CI [XX, XX], $p = XX$, effect size XX), and first-keystroke latency by XX ms (95% CI [XX, XX], $p = XX$, effect size XX); CPM and attempted characters gave $p = XX$ and XX, respectively. Table 10 and Fig. 8 summarize the subjective ratings: Friedman tests yielded $\chi^2 = XX$ ($p = XX$) for readability, $\chi^2 = XX$ ($p = XX$) for stability, and $\chi^2 = XX$ ($p = XX$) for immediate visual comfort; Holm-corrected pairwise comparisons between the deployed full configuration and the unmasked anchor gave $p = XX$, XX, and XX on the three items, respectively.

VI. DISCUSSION

A. EVIDENCE HIERARCHY AND CLAIM SCOPE

The most directly supported conclusion is: within a single-UVC-camera feasibility study, temporal pixel masking substantially reduces conventional OCR character recovery and exact-match rates from calibrated short-exposure screen photographs under the tested S600 camera, display, corpus, and

9 geometric conditions. This is not a general anti-photography claim. It is a controlled UVC, single-frame claim about bulk or unattended conventional-OCR collection, where each target is represented by one isolated fixed-exposure image. The value of this narrow claim is that it converts a previously plausible display-side idea into a measured operating window with named failure modes. The 10,575 real-capture images reflect component, parameter, and attack-condition coverage within this UVC link. Because the corpus is unbalanced and covers one webcam, the total sample count is not cross-device or cross-scenario inference evidence. Supplementary detection/tracking diagnostics, VLM probes, and simulations respectively answer whether the degradation may transfer to other visual tasks, whether stronger recognizers break through, and where the full-cycle integration boundary lies; together they define boundaries rather than extend the primary claim. The passive baselines are software proxies with specified pixel-domain outputs, not physically or perceptually matched products. The present data therefore do not support comparative superiority over physical privacy filters, prior temporal-display systems, or differently tuned static obfuscation.

B. THE MATHEMATICAL TENSION OF LINEAR INTEGRATION

The scheme exploits a difference between human temporal perception and a short camera exposure. Its mathematical boundary, however, is defined in an ideal linear display-camera radiance model rather than by treating human vision as a linear summation operator. In that model, complementary subframes can be re-summed to recover the original ($\sum I_k = I$), so a camera that accumulates a registered full cycle defeats the linear decomposition. Research on VLM adversarial robustness [45] further shows that large models extract partial semantic information from perturbed images; the probes in §V-C confirm this. VLMs recover large-font content from single short-exposure photographs and obtain stronger signals from the 150-frame video mean and long-exposure views. The archived digital evaluation reaches 95.2% under the per-sample strongest attack across all evaluated reconstructions; pure full-cycle temporal-mean recovery reaches 94.3% (§V-E2).

C. PHYSICAL-LAYER CONFOUNDS AND THE SIMULATION-TO-REAL GAP

Two physical-layer mechanisms systematically separate real-capture results from idealized single-subframe simulation.

Direction 1: real short-exposure recovery exceeds simulation (favoring the attacker). Simulated single-subframe OCR yields 0–2.6% character recovery (Table 8), whereas real-capture short-exposure recovery under comparable masking is 19.2% (mask-only, Table 4). Two physical effects could contribute. First, rolling-shutter row mixing can make different rows sample different display phases, although the S600 readout time and phase were not measured. Second, LCD response transitions can blend adjacent subframes; the

AOC 24G51Z's specified 1 ms GtG already occupies about 24% of a 4.17 ms slot. The real-capture measurements include both effects, but this experiment does not identify their separate contributions. The simulation is therefore an idealized single-subframe reference, not a physical lower bound.

Direction 2: real integration recovery is lower than simulation (favoring the defender). The per-sample strongest digital attack recovers 95.2% (§V-E2), whereas the real-capture 150-frame video mean yields 71.1% for deployed and 47.9% for high-suppression in the full pool. After the underexposed position is excluded, these increase to 79.7% and 53.9%, respectively. Camera ISP nonlinearities (gamma, white balance, noise reduction), moiré interference between the display pixel grid and the sensor Bayer pattern, optical blur from defocus and lens aberration, and imperfect temporal registration across frames all degrade the attacker's integration quality. Even the more conservative sensitivity values remain below the digital strongest-attack result, but the gap is smaller than the pooled figures suggest; a purpose-built attack setup could narrow it further.

These two directions imply that the simulation and real-capture experiments answer different questions: simulated single-subframe recovery characterizes the algorithm's theoretical information hiding, while real-capture recovery characterizes the end-to-end system including panel, camera, and ISP confounds.

D. ROLE AND COST OF THE HIGH-SUPPRESSION PROFILE

The high-suppression profile breaks strict completeness through stripe and glyph perturbations, reducing integration recovery at the cost of worsened reconstruction metrics. In the digital pipeline, its full-cycle SSIM is 0.763 and ΔE_{00} is 4.62, far below the deployed profile's 0.948 and 1.46, respectively. In the full real-capture pool it achieves 0% short-exposure exact match and 5.6% exact match under the 150-frame video mean, while video character recovery remains 47.9%; after excluding the underexposed position, short-exposure character recovery is 5.6% and video character recovery is 53.9% (§V-B). It therefore misses the $< 5\%$ short-exposure target and only mitigates integration leakage. Whether users accept the visible stripes and grain remains untested: the study in §V-F covers only the deployed profile and core mask conditions.

This profile split is central to deployment interpretation. The deployed profile is the only profile connected to the planned readability and comfort study, but it is also the weaker protection setting under sensitive-token and integration attacks. The high-suppression profile is closer to the security objective but lacks human-subject acceptability evidence. The present manuscript therefore supports a feasibility niche and an evidence hierarchy, not a completed deployment recommendation.

E. VLM ATTACKERS AND COUNTERMEASURE DIRECTIONS

Protection failure concentrates on large-font short snippets (§V-C), and temporal averaging further amplifies the sig-

nal: glyph contour fragments in a single subframe suffice for language-prior completion, and multi-cycle coverage approaches the original. We retain this negative result as boundary evidence because it directly constrains the method's deployability: the scheme mitigates conventional OCR short-exposure capture but does not resist modern VLM screen-reading. Three unvalidated research hypotheses follow: (1) **font-size-adaptive mask granularity**, scaling the mask cell with stroke width so that a single subframe no longer preserves character topology; (2) **glyph decoys**, testing whether false strokes can reduce single-frame VLM completion; and (3) **field-level content policies**, rendering high-risk fields in dense small font or enabling the high-suppression profile per field. The decoy idea cannot be presented as a solution within the current $\sum_k \mathbf{N}_k = \mathbf{0}$ framework: any exactly complementary decoy also cancels under full-cycle integration and inherits the same linear boundary. None of these directions has been experimentally validated, and adversarial perturbations targeting VLM encoders are further constrained by digital-to-physical transferability (§II). More generally, any information visible to the observer within the integration window is in principle recoverable by a full-cycle camera paired with a strong-prior model; VLMs merely reduce the fragment count needed. The defense objective should shift from "unreadable to any attacker" toward increasing attack cost and reducing unattended bulk collection.

F. SUSTAINED VIDEO ATTACKER

The evaluated video attack does not establish a four-frame bypass. The capture pipeline records 150 short-exposure frames at nominal 60 fps, and the reported temporal-mean image averages the complete burst, corresponding to approximately 2.5 s of nominal capture. This attack does not require the mask seed, phase, or precise frame selection because sustained averaging accumulates multiple display cycles. Its pooled recovery is 71.1% for deployed and 47.9% for high-suppression, rising to 79.7% and 53.9% after excluding the underexposed position. The archive also contains aggregate views selected from the same full burst, but they do not establish recovery from an independently acquired short window. A duration-versus-recovery curve and pre-specified short-window analysis remain unevaluated. The single-shot scope should therefore be distinguished from the measured sustained-recording boundary without assigning an unsupported minimum bypass duration.

G. DEPLOYMENT PATH

Real-world deployment would require OS driver-level framebuffer injection (DXGI / KMS-DRM / CoreDisplay), panel/backlight timing coordination, and high-refresh panels meeting the target cycle rate. These steps introduce color management, luminance saturation, frame drops, and synchronization errors that could alter algorithm behavior; they cannot be treated as engineering packaging that leaves conclusions unchanged. The deployed profile's 96.5% sensitive-token recovery under long exposure means credentials and

similar high-risk fields should not rely on it alone. Such fields should use the high-suppression profile or dense small-font rendering under a field-level policy, with the caveat that these options impose unresolved readability costs and do not close the video/VLM boundary.

H. ETHICS STATEMENT

This research is defensive in nature, aiming to protect screen content from unauthorized photography. The attack analysis (§V-E2) is intended for honest assessment of security boundaries, not as an attack guide. The planned user study would involve human subjects in minimal-risk research; no participant data have been collected. Recruitment will begin only after the competent institutional or departmental authority issues a written approval, exemption, or waiver determination. The protocol specifies informed consent, photosensitivity screening, and voluntary withdrawal as described in §V-F.

I. LIMITATIONS

- 1) The user study (§V-F) is a protocol, not completed evidence: no participant data exist yet, so the deployed profile's usability remains unvalidated. Even when conducted in a controlled 240 Hz laboratory, external validity will require broader participant samples and display conditions.
- 2) Immediate visual comfort is a single-item scale measured after short exposure and does not extend to long-term fatigue or clinical visual discomfort.
- 3) The PoC is application-layer only, without OS drivers or hardware backlight. Luminance, HDR, and color-difference conclusions come from digital models, lacking photometric or panel-response measurements. Actual playback slot timing was not verified with a photodiode, high-speed camera, or oscilloscope. OS compositor jitter, VSync mismatch, and dropped frames can shift subframe boundaries unpredictably; if a short-exposure window straddles the intended slot edge, the camera captures cross-slot content and real recovery may exceed the values reported here. Because the experiment lacks a luminance-matched static control and matched noise-off enhanced profiles, it cannot separate temporal sparsity, duty-cycle luminance, static overlays, panel response, camera exposure, and ISP processing as mechanisms.
- 4) Real-capture OCR used one S600 UVC webcam; ambient illuminance was not recorded; exposure values have two calibration sets. The unbalanced design with repeated content means bootstrap intervals do not substitute for cross-device replication. Uncontrolled ambient light variation may affect results through two paths: (a) the camera's auto-exposure, auto-white-balance, and auto-gain algorithms respond to ambient conditions, altering effective exposure on the screen content; (b) ambient light reflected off the screen surface changes the captured signal-to-noise ratio. If geometric configurations were captured at different times

of day, systematic ambient variation could confound geometric-factor analysis. Increased reflection or sensor noise would most likely suppress the already weak protected signal more than the unprotected baseline, biasing protected recovery downward and favoring the defense; automatic gain or exposure compensation could act in the opposite direction. Because illuminance was not logged, the net direction and magnitude cannot be estimated. This uncertainty primarily weakens cross-geometry comparisons; the large within-link short-exposure reduction remains directly observed but is not immune to the confound.

- 5) Real-capture detection used 150 COCO images at a single position; tracking used 450 still-frame captures from one MOT17 sequence, not continuous video. Simulated detection used 8 images with approximate metrics.
- 6) VLM evaluation covers one camera, 0° on-axis, one profile, and three commercial API models (12 video segments per cell). Models may update over time. Public document content may inflate recovery via training-data memorization. We did not evaluate multi-cycle adaptive reconstruction, rolling-shutter phase search, or pre-masking malware interception. Full-cycle integration remains the fundamental boundary.
- 7) Conventional OCR evaluation covers Surya, EasyOCR, and Tesseract but no commercial OCR API. Best-of-engine over these three systems is attacker-favorable within the evaluated set, yet Google Cloud Vision, Azure AI Vision, Baidu OCR, or future recognizers may obtain higher recovery.
- 8) Baseline comparisons use unmatched software proxies rather than physically tested privacy filters or prior temporal-display systems. They contextualize the observed trade-off but do not establish comparative superiority over those alternatives.
- 9) The deployed profile is not suitable as the sole protection layer for sensitive fields: complete sensitive-token recovery remains 24.0% under short exposure and 96.5% under long exposure. Field-level masking, independent access controls, and exposure to the video/VLM boundary must be considered together.
- 10) All 12 test content items are English/ASCII text. CJK characters (Chinese, Japanese, Korean) have much higher stroke density than Latin letters; at the same mask granularity, CJK strokes are more likely to be disrupted (stroke widths closer to mask cell size), but higher inter-character visual similarity may also increase OCR confusion. The net effect on protection efficacy for non-Latin scripts is unknown and left to future work.

VII. CONCLUSION

We implemented and evaluated temporal pixel masking as a controlled UVC feasibility study for single-frame conventional-OCR mitigation. In 10,575 real-capture images

from one UVC webcam, best-of-engine OCR character recovery falls from 94.1% to 15.1% for the deployed profile and 5.0% for high-suppression under calibrated short exposure, with exact match falling to 0.4% and 0%. Excluding one underexposed geometry raises the protected estimates to 16.7% and 5.6%, so high-suppression remains near but does not meet its strict $< 5\%$ target. In matched content-cluster contrasts, unprotected short exposure exceeds deployed by 78.0 percentage points (95% CI [75.5, 80.3]) and high-suppression by 89.1 percentage points ([85.0, 92.4]). This result is the paper's central contribution: it defines, measures, and bounds a measured operating window for disrupting unattended conventional-OCR collection from one isolated fixed-exposure UVC frame per target. Such a collector must accept substantially poorer output, sustain multi-frame capture, or use a stronger recognizer. Sensitive-token recovery remains 24.0% under deployed short exposure, so the window does not extend to all sensitive fields.

The measured bypass boundary is equally important. A temporal mean over 150 short-exposure frames, nominally 2.5 s at 60 fps, recovers 71.1% character for deployed and 47.9% for high-suppression; excluding the underexposed geometry raises these values to 79.7% and 53.9%. Under long exposure, deployed recovery is 60.9% versus 47.3% unprotected, so temporal masking does not provide a general integration defense. In the tested on-axis scan, Qwen3-VL exactly recovers 28/36 captures from 12 content items at 0.5 m under high-suppression short exposure. The content-type character analysis separately shows that failures concentrate in large-font short snippets. Sustained video aggregation and modern VLMs therefore sit outside the measured operating window as practical bypasses. The minimum video duration required for comparable recovery remains unmeasured.

These results show that temporal pixel masking is not a general anti-camera defense, but it can materially degrade unattended conventional-OCR batch collection from isolated calibrated short-exposure UVC captures. The main value of the study is therefore an evidence-bounded protection window, plus a set of measured failure boundaries that future display-side privacy systems should treat as design constraints. Concrete next steps are to close the profile split by testing usability for stronger suppression settings, evaluate font-size-adaptive mask granularity and field-level policies against VLMs, capture matched noise-off enhanced profiles, and validate cross-device behavior on physical systems.

DATA AND CODE AVAILABILITY

The software, experimental configurations, de-identified per-sample OCR results, VLM probe records, and figure-generation scripts are maintained in the project repository at <https://github.com/docandyc/privacy-display>. An immutable release aligned with the final submitted manuscript, figures, analysis scripts, and de-identified records will be minted before or with the submission package; no release identifier is claimed here. The canonical archived identifier `capture_hardened` and the legacy `v1m` tag both denote

the paper's high-suppression configuration (§IV-E). If the user study is conducted, its database `study_formal.db` (containing student IDs and names) will not be released; public materials will include only de-identified results, exclusion audits, and analysis scripts.

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[PLACEHOLDER: Acknowledgment to be filled.]

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